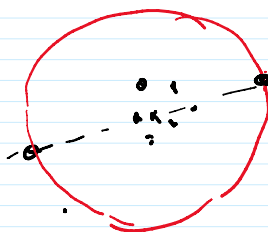
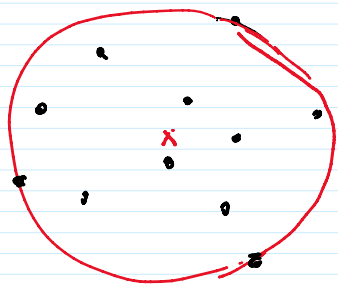


Problem

Given set S of n points in 2D, compute the min circle enclosing S .



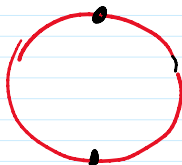
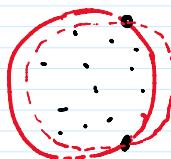
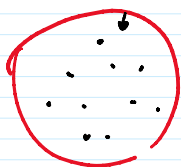
History:

- convex hull?
- "Voronoi diagram" $O(n \log n)$ time in 2D
- Megiddo / Dyer '83: $O(n)$ time !! but complicated...
- Seidel / Welzl '91: $O(n)$ expected time

Obs C^* is unique & has 3 pts on boundary or 2 pts on boundary as "diametrical pair"

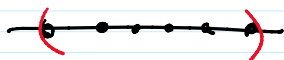
$\Rightarrow$ brute force $O(n^3 \cdot n) = O(n^4)$ time algm

Pf Sketch: By contradiction.



□

Digression:



Subproblem (1D) Given n numbers $S = \{x_1, \dots, x_n\}$, find the min.

Standard "Incremental" Alg'm:

- randomly permute $x_1, \dots, x_n$
- ans = ∞
- for $i = 1$ to n
- if $x_i < \text{ans}$ then
ans = x_i ← (*)

Q: how many switches (*)?
expected #

rewrite alg'm:

min(S): // "backwards" version

0. if $S = \emptyset$ return ∞
1. pick $x \in S$ randomly
2. $\text{ans} = \min(S - \{x\})$
3. if $x < \text{ans}$ then
- 4. $\text{ans} = x \leftarrow (*)$
5. return ans

$$\Pr[(*) \text{ is done}] = \Pr[x = \min(S)] \\ = \frac{1}{n}.$$

$$\Rightarrow T(n) = T(n-1) + \frac{1}{n} \quad \text{by linearity of expectation}$$

$$\Rightarrow T(n) = \frac{1}{n} + \frac{1}{n-1} + \frac{1}{n-2} + \dots + \frac{1}{2} + 1 \\ = n^{\text{th}} \text{ Harmonic number} \\ = \ln n + O(1) \\ = O(\log n).$$

Seidel/Welzl's Rand. Alg'm ('91) (Las Vegas)

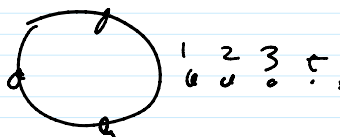
idea - incrementally add pts **RANDOMLY**

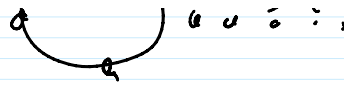
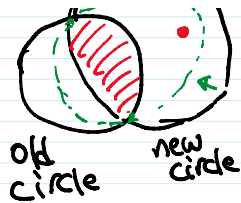
if new pt p is **inside** circle
unchanged ✓

else new circle must pass
thru p on boundary



Pf "Sketch":

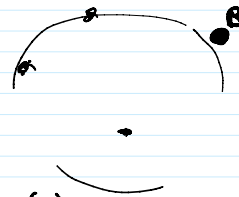




circle(S, B):

// find min circle enclosing S with extra constraint
that boundary passes thru pts in B

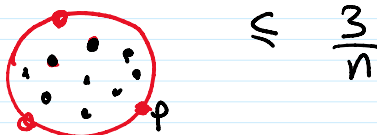
0. if $S = \emptyset$ or $|B| = 3$ return ---
1. pick $p \in S$ randomly
- 2. $C = \text{circle}(S - \{p\}, B)$
3. if p outside C then
4. $C = \text{circle}(S - \{p\}, B \cup \{p\})$ (*)
5. return C .



Rand. Analysis:

let $T_b(n) =$ expected runtime for $|S|=n, |B|=b$.
for any fixed S, B ,

$$\Pr[(*) \text{ is done}] = \Pr[p \text{ is on boundary of circle}(S, B)].$$



$$T_b(n) \leq T_b(n-1) + \frac{3}{n} \cdot T_{b+1}(n-1) + O(1)$$

base case: $T_3(n) = O(1)$.

$$T_2(n) \leq T_2(n-1) + \frac{3}{n} \cdot O(1) + O(1)$$

$$\Rightarrow T_2(n) = O(n)$$

$$T_1(n) \leq T_1(n-1) + \frac{3}{n} \cdot O(1) + O(1)$$

$$\Rightarrow T_1(n) = O(n)$$

$$T_0(n) \leq T_0(n-1) + \frac{3}{n} O(n) + O(1)$$

$$\Rightarrow T_0(n) = \boxed{O(n)} \text{ expected time}$$

Rmk: in d dims, $O(\frac{(d+1)!}{\text{const}} n)$ expected time

(Megiddo: $O(2^{2^d} n)$)

Problem Given n numbers,
find median $((n/2)^{\text{th}}$ smallest)

73, 81, 60, 52, 43, 92, 100,

Sorting: $O(n \log n)$ time
better?

$1 \cdot n \log n + O(n)$ comps.

Known deterministic algm:

Blum, Floyd, Pratt, Rivest, Tarjan '73:

$O(n)$ time "median-of-medians-of-5"

$\sim 16n$ comparisons

$\sim 5.4n$ "

Schönhage, Paterson, Pippenger '76 $\sim 3n$ comps

Dor, Zwick '95 $\sim 2.95n$

Lower bds:

Bent, John '80 $\sim 2n$

Dor, Zwick '96 $\sim 2.000\,000\,000\,000\,000\,000\,001\,n$

Randomized algms?

quickselect:

pick rand pivot

quickselect:

pick rand pivot

recurse on one side

$\Rightarrow O(n)$ expected time

$\hookrightarrow$
if careful, $\leq 3.4n$ comps expected